Venkata Lakshmi Parimala Pasupuleti

Software Engineer · Python · Django · PostgreSQL · Redis · Kafka · AWS · AI-First · Distributed Systems · Full SDLC · FastAPI

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PROFESSIONAL SUMMARY

Software Engineer with 8+ years building production backend systems at scale. At Entrata, owned 20+ full-stack modules end-to-end — Python, PHP, Java, PostgreSQL, Redis, RabbitMQ, AWS — for 100K+ monthly active users at 99.5% uptime across 120+ Agile sprints. Self-starter who thrives without a spec: owned a business-critical compliance pipeline solo for 7 years, no project manager, no playbook. AI-first: built AgentProbe (open-source AI eval toolkit with CI regression harness and trace-level observability) and LeaseI (8-agent LangGraph platform with 93.6% precision across 1,240 test cases) because the tooling didn't exist and the problem needed solving. M.S. Data Analytics Engineering, George Mason University (GPA 3.87). — github.com/venkatalakshmiparimala/AgentProbe

WORK EXPERIENCE

Blue Horse Digital Sep 2026 – Present

AI/ML Engineer (Volunteer) · Remote

- Building ML models and data APIs; integrating AI components across backend and reporting layers; debugging cross-stack reliability issues.

George Mason University Aug 2024 – Aug 2025

Research Associate + Teaching Assistant

- Built multi-source ML prediction pipelines (Python, R) across 190+ countries (R^2=0.977); automated quality checks (Isolation Forest); HuggingFace fine-tuning + inference API deployment; statistical validation (hypothesis testing, confidence intervals). Workshops to 600+ students across Python and SQL.

Entrata (PropTech / Real Estate SaaS) Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- **Full-stack backend ownership across 20+ production modules** (Python, PHP, Django, Java, PostgreSQL, Redis, RabbitMQ, AWS) serving 100K+ monthly active users at 99.5% uptime — designed, built, tested, deployed, and operated end-to-end across 120+ Agile sprints. Owned a mission-critical compliance pipeline solo for 7 years with no spec, no PM, no playbook. Zero compliance violations. Zero security incidents.
- **Distributed systems built from first principles** — event-driven architecture (RabbitMQ/Kafka) with retry logic and idempotency guarantees; multi-tenant PostgreSQL schema design for 100K+ units; 200+ SQL optimizations (30% performance gain); Docker microservices redesign (50% scalability improvement, 40% deployment time reduction). Owned every production incident through to permanent root-cause fix.
- **AI-first before it was a policy** — instrumented systems with OpenTelemetry + Prometheus + Grafana (40% reduction in debugging time); built Python/Bash automation eliminating 80% of manual operational workload; mentored 5+ engineers; 300+ peer code reviews.

Entrata India, Pune Jul 2017 – Sep 2018

Associate Software Engineer

- Built backend modules (PHP, Python, Java, PostgreSQL, Django); led AWS S3 migration; data pipelines across hundreds of thousands of records.

PROJECTS

AgentProbe: AI Evaluation Infrastructure CI-INTEGRATED OPEN SOURCE

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · OpenTelemetry | github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI backend; 5-metric eval engine; LLM-as-judge scoring; CI regression harness with golden datasets; distributed model routing across OpenAI/Anthropic/Groq/Ollama. Built because the tooling didn't exist - the same AI-first, high-agency instinct Nextdoor describes.

LeaseI: 8-Agent AI Platform LANGGRAPH 93.6% PRECISION

Python · LangGraph · FastAPI · SQLite · OpenAI API | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Distributed 8-agent workflow with durable state management, fault-tolerant HITL escalation, and RAG over document corpus. 93.6% precision, 87.9% F1 across 1,240 test cases. LLM reasons; deterministic logic decides every regulated outcome.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur 2017

SKILLS

Languages & Frameworks: Python (8+ yrs), Django, PHP, Java (JVM / Spring Boot), JavaScript, TypeScript, SQL, PL/SQL, Bash — FastAPI, React.js, REST APIs

Backend & Data: PostgreSQL, Redis, Apache Kafka, RabbitMQ, Microservices, Distributed Systems, Event-Driven Architecture, Multi-Tenant Architecture, Idempotency, Docker, Kubernetes, AWS (S3, EC2, RDS, Lambda, Step Functions)

AI & Observability: LangGraph, LangChain, RAG Pipelines, LLM Evaluation, AgentProbe, Agentic Workflows, OpenTelemetry, Prometheus, Grafana, CI/CD (GitHub Actions), Terraform, Performance Engineering, PyTorch, HuggingFace